

Michael Plessner

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[michaelplessner.github.io](https://github.com/michaelplessner)

Education

Pennsylvania State University (PSU) - PhD in Physics	2022 - 2027
<i>John Randall Shuman Troxell Memorial Scholarship in Physics</i>	2025
<i>Edward A. and Rosemary A. Mebus Graduate Fellowship in Physics</i>	2024
<i>Edward M. Frymoyer Honors Scholarship</i>	2024
Pennsylvania State University (PSU) - MSc in Physics	2020 - 2022
<i>Paul Berg and Daniel J. Larson distinguished graduate fellowship</i>	2020
The University of Hong Kong (HKU) - Semester Abroad	2016
Northeastern University (NEU) - BSc in Physics with Honors	2015 - 2019
<i>Merit scholarship for four years, also awarded a Dean's scholarship</i>	
<i>Presidential global scholarship to perform research in Italy</i>	
<i>Presidential global scholarship to study in Hong Kong</i>	

Research

PSU – PhD Supervisor: Dr Jacob Bourjaily 2022 - 2027

Broadly interested in the mathematical structures of QFTs and improving “on-shell” methods.

Ongoing research programs include,

*Colour in gauge theory, applications of classical and quantum representation theory,
Covariant formalisms for massive particles and generalized recursion relations,
Mathematica packages for handling the math and physics of scattering,
Generalized double copy and colour-kinematics duality,
Combinatoric models of amplitudes.*

NEU – Supervisor: Dr James Halverson 2019 - 2020

Studied solvability of Diophantines in string index calculations under topological transitions.

CERN / CMS / NEU – Supervisor: Dr Toyoko Orimoto 2018 - 2019

Used ROOT to perform test beam timing resolution analyses on HL-LHC ECAL hardware.

Università degli Studi di Milano – Supervisor: Dr Marcel Di Vece 2017 - 2018

Modeled “planetary” nanoparticle systems formed by a sputtering cluster source in Python.

CERN / ATLAS / HKU – Supervisor: Dr Yanjun Tu 2016 - 2017

Used ROOT to simulate backgrounds and search for SUSY in dilepton events at CERN.

Publications and Preprints

1. S. Caron-Huot, J. Fei, **M. Plesser**, *Canonical Bases for Classical Group Tensor Products*, In Preparation
2. J. Bourjaily, **M. Plesser**, *Covariant BCFW for all Masses*, In Preparation
3. J. Bourjaily, **M. Plesser**, P. Velie, *Constructing Concrete Colour Tensors for Simple Lie Algebras*, [2609.13365](#)
4. J. Bourjaily, **M. Plesser**, P. Velie, *Tree Amplitudes in Pure Gauge Theory with Charged Matter*, [2604.18672](#)
5. J. Bourjaily, **M. Plesser**, C. Vergu, *The Colour Dependence of Amplitudes*, **JHEP** 08 (2026) – [2512.23806](#)
6. J. Bourjaily, **M. Plesser**, C. Vergu, *The Many Colours of Amplitudes*, [2412.21189](#)
7. J. Bourjaily, **M. Plesser**, et al, *Gauge-Invariant Double-Copies via Recursion*, **Phys. Rev. Lett.** 131 (2023) – [2307.02542](#)
8. J. Halverson, **M. Plesser**, F. Ruehle, J. Tian, *Kahler Moduli Stabilization and the Propagation of Decidability*, **Phys. Rev. D** 101 (2020) – [1911.07835](#)
9. S. Marom, **M. Plesser**, et al, 'Planetary' silver nanoparticles originating from a magnetron sputter plasma, **J. Phys. D: Appl. Phys** 52 (2019)

Teaching Assistantships at Penn State University

Introductory Mechanics (PHYS 211)	2024, 2026
Introductory Electromagnetism (PHYS 212)	2022, 2023, 2024
Thermal Physics (PHYS 213)	2022
Wave Physics (PHYS 214)	2022, 2025
Algebra-Based Introduction to Mechanics (PHYS 250)	2025
Electromagnetism (PHYS 400)	2021
Subatomic Physics (PHYS 406)	2022, 2025
Quantum Mechanics I (PHYS 410)	2021, 2025
Graduate Quantum Mechanics II (PHYS 562)	2022
Computational Resources and Curriculum Development	2026

Invited Talks and Visits

Invited talk at Penn State's Math department (upcoming)

"Canonical bases for quantum groups and applications to physics"

Invited to visit McGill University by Simon Caron-Huot (upcoming)

Invited to visit the IAS by Nick Early and Nima Arkani-Hamed IAS, NJ, USA, 2026

Invited talk at UCT Applied Maths department UCT, Cape Town, SA, 2025

"What can representation theory teach us about gauge theory?"

Invited talk at April APS Conference Denver, CO, USA 2019

"Test beam studies of precision timing with the upgraded CMS-ECAL"

Conferences, Summer Schools, and Workshops

AI in CAmplitudes UCLA, CA, USA (upcoming)

Loop-The-Loop 3 Workshop on Amplitudes (Online), (upcoming)

Prospects in Theoretical Physics (PiTP) Summer School IAS, NJ, USA, 2026

Gong show: "QED from $N=4$ "

Amplitudes Summer School U. of Southampton, UK, 2026

Symbology@15 Conference MPP, Munich, DE, 2025

Gong show: "Masses, flavours, and colours in amplitudes"

TASI Summer School CU Boulder, CO, USA, 2025

Gong show: "Colour structures and amplitudes"

Michigan Interdisciplinary Meeting on Amplitudes U. of Michigan, MI, USA, 2025

Gong show: "Graph games from representation theory"

Loop-The-Loop 2 Workshop on Amplitudes (Online), 2025

Strings and Geometry Conference ICTP Trieste, IT, 2025

Loop-The-Loop Workshop on Amplitudes (Online), 2024

Mathematical Aspects of Scattering Amplitudes Conference GGI, Florence, IT, 2024

Amplitudes Conference and Summer School IAS, NJ, USA, 2024

Gong show: "A ridiculous formula for the number of $SU(N)$ gluonic colour factors"

Amplitudes Conference and Summer School CERN, FR / CH, 2023

Gong show: "Covariant BCFW for all masses"

Amplitudes Conference and Summer School Charles University, Prague, CZ, 2022

The Amplitude Games Summer School MITP (Online), 2021

Outreach and Mentorship

- 1) Provided tutoring and career advice to minority students at the HBCU (historically black college / university) North Carolina Central University as part of the PREM (Partnership for Research and Education) program

Graduate Electromagnetism

2024, 2025

Undergrad Electromagnetism and Quantum Mechanics

2023, 2024, 2025
- 2) Peer supervisor for junior PhD students working in Dr. Bourjaily's group at Penn State

Philip Velie

2023-2027

Malek Telfah

2026-2027
- 3) Tutored Penn State physics PhD students remediating first-year courses 2025
- 4) Research talks at Penn State's "Primordial Universe and Gravity" (PUG) seminar

"TBD" (upcoming)

2026

"Why (Non-Abelian) Gauge Theory?"

2025

"Correcting Coulomb and Newton with On-Shell Methods"

2024

"Quantum Field Theory Without Fear"

2023
- 5) Research talk to the physics and chemistry club at State Area High School

"A Brief History of QFT and Why the Sky is Blue"

2022
- 6) Research talk at Penn State's "Mathematical Aspects of Physics" (MAP) seminar

"Why Good Variables Matter: Representations of Poincaré"

2022

Rock Climbing

- Semi-professionally sponsored by Mad Rock Climbing from 2014 - present.
- Commercial Routesetter with over a decade of experience.
 - *Routes:*
Salathé Wall on **El Capitan** (5.9 C2, 3500' Yosemite, USA),
Snake Dike on **Half Dome** (5.7 R, 2000' Yosemite, USA),
Time Wave Zero on **El Toro** (5.12a, 2300' El Potrero Chico, Mexico),
Flatirons scramble during the TASI 2025 summer school in Boulder, CO
El Camino Royale 500' - **Royal Arch** 100' - **Dodge Block** 700'
 - *Boulders:*
Mooiste Meisie (v13 / 8B, South Africa),
Donkeystrength (v12 / 8A+ FA, South Africa),
Freak Brothers (v11 / 8A, Switzerland),
Monstrous Moonshine (v11 / 8A FA, NC, USA),
973 to the Dangerzone (v6 / 7A R FA, Hong Kong)